

Evaluating Credit Card Minimum Payment Restrictions

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The views presented are those of the authors and do not necessarily reflect those of the Bank of Canada

Research Question

What is the magnitude of the trade-offs of a hard paternalistic policy that restricts consumer choices?

Contributions

- Behavioral Public Economics:
 - Evaluating hard paternalism
- Credit Cards & Consumer Financial Protection Policy:
 - Equilibrium responses to a market-wide credit card policy

1. Motivation
2. Quebec Policy
3. Data & Methodology
4. Effects on Consumers
5. Supply Responses

Motivation

Credit Card Policy Trade-Off

Imagine two types of consumers use credit cards: (Heidheus & Köszegi, 10, 15; Campbell, 16)

- **“Sophisticates”** optimize
- **“Naïfs”** overborrow (e.g., financially unsophisticated, naïve present focus, targeting)

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Unfortunately, “ideal policies” of information & nudges ineffective (e.g., Agarwal et al. 15; Keys & Wang, 15; Seira et al., 17; Adams et al. 22; Batista et al., 24; Guttman-Kenney et al., 25)

→ “**nudges are not enough**” (Laibson, 20)

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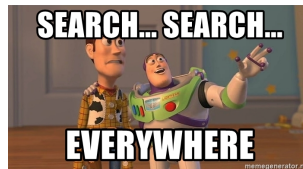
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**Need to evaluate hard paternalistic policies trade-off benefits of 1 vs. costs of 2
& consider supply responses**

Hard Paternalistic Policy To ↓ Credit Card Debt?

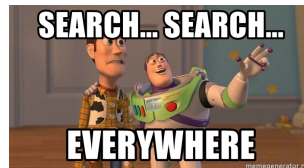
Mandating Tighter (Higher) Credit Card Minimum Payments

- Want policy forcing credit card lenders involuntarily ↑ minimum payment



Where to Study Tighter Minimum Payments Policy

- Want policy forcing credit card lenders involuntarily $\uparrow$ minimum payment
- Policy variation does not exist in U.S.
- Prior research studied lenders *voluntarily* changing minimum payments (D'Astous & Shore, 17; Keys & Wang, 19, Castellanos et al., 24; Medina & Negrin, 22), & Turkish policy effectively preventing using credit cards for borrowing (Agarwal et al., 23)
- **Takes us to Canada!**



Quebec Policy

Quebec is a Laboratory for Hard Paternalism Aiming to ↓ Credit Card Debt

Quebec office de la protection du consommateur:

*“These changes are part of the measures adopted by the government
to prevent indebtedness”*

**“It is advantageous to pay the balance of your credit card each month,
*because no credit charges are then applicable”***

Quebec rolls out new credit card rules aimed at lowering high household debt

By Reuters Staff

2 MIN READ



OTTAWA, Aug 2 (Reuters) - The Canadian province of Quebec has become the first in the country to require minimum credit card payments as legislators try to tackle rising household debt levels.

Quebec introduces credit card regulations aimed at curbing consumer debt

CTVNews.ca Staff



Ben Cousins CTVNews.ca Writer
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Published Thursday, August 1, 2019 4:38PM EDT

No credit card minimum payment requirements:

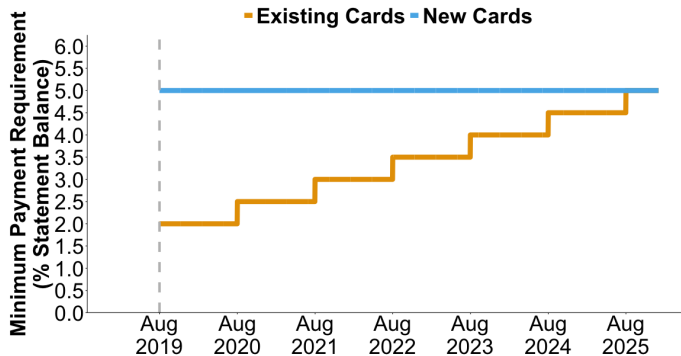
- Pre-August 2019: All Canadian provinces
- Post-August 2019: All Canadian provinces **except Quebec**

Quebec Policy Experiments

No credit card minimum payment requirements:

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Quebec 2019 credit card minimum payment policy phases:



Data & Methodology

Consumer Credit Reporting Data

Bank of Canada's anonymized TransUnion consumer credit reporting data

- January 2018 to November 2024
- Monthly, tradeline-level information on each credit card held in Canada
- Includes account dates, balances, credit limits, minimum payments, actual payments

(See Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang, 2025 *JEL* guidance!)

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Mintel CompereMedia Data

- Credit card offers received via mail/email by 8,000 Canadian households
- Includes contract terms (e.g., APRs, credit limits)
- Measure of supply (e.g., Ru & Schoar, 2020; Grodzicki, 2022, Han, Keys, & Li 2018)

Synthetic Difference-in-Differences (SDID) Methodology

Arkhangelsky, Athey, Hirshberg, Imbens & Wager (2021 AER); Arkhangelsky & Imbens (2024):
- SDID outperforms both synthetic controls & difference-in-differences approaches

Follow Arkhangelsky et al. (2021) estimating:

$$(\hat{\delta}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\delta, \mu, \alpha, \beta}{\operatorname{argmin}} \left\{ \sum_{p=1}^9 \sum_{t=-18}^6 (y_{p,t} - \mu - \alpha_p - \beta_t - \delta(\mathbb{I}\{t \geq 0\}_t \times \mathbb{I}\{QUEBEC\}_p))^2 \hat{\omega}_p \hat{\lambda}_t \right\} \quad (1)$$

- Outcomes in Quebec (Q) are compared to “Synthetic Quebec” (SQ) of 8 provinces
- Weights using statement balances $t = -18$ to $t = -13$ to allow for anticipation effects

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- Weights using statement balances $t = -18$ to $t = -13$ to allow for anticipation effects
- Use same weights for all outcomes (weights range from 0.11 to 0.14)
- Standard errors using placebo variance estimation to account for single treated cluster
- Show dynamic effects relative to pre-policy baseline ($t = -18$ to $t = -13$):

$$\delta_{\tau} = \left(\bar{y}_t^Q - \bar{y}_t^{SQ} \right) - \left(\bar{y}_{baseline}^Q - \bar{y}_{baseline}^{SQ} \right) \quad (2)$$

Effects on Consumers

How Lenders Implemented First Policy Phase in August 2019?

Non-binding, Bind, Bind and Go Beyond

Lender	Existing Cards Opened Before August 2019			
	July 2019		August 2019	
	Non-Quebec	Quebec	Non-Quebec	Quebec
A,B	\$10 + interest + fees		max {2.0% statement balance, \$10}	
C,D,E	\$10 + interest + fees		max {2.5% statement balance, \$10}	
F	\$10 + interest + fees		max {3.0% statement balance, \$10}	
G	max {1.0% statement balance, \$10}		max {3.0% statement balance, \$10}	
H,I	max {2.0% statement balance, \$10}		max {2.0% statement balance, \$10}	
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After 2019, lenders typically $\uparrow$ 0.5 percentage points each year until reaches 5%

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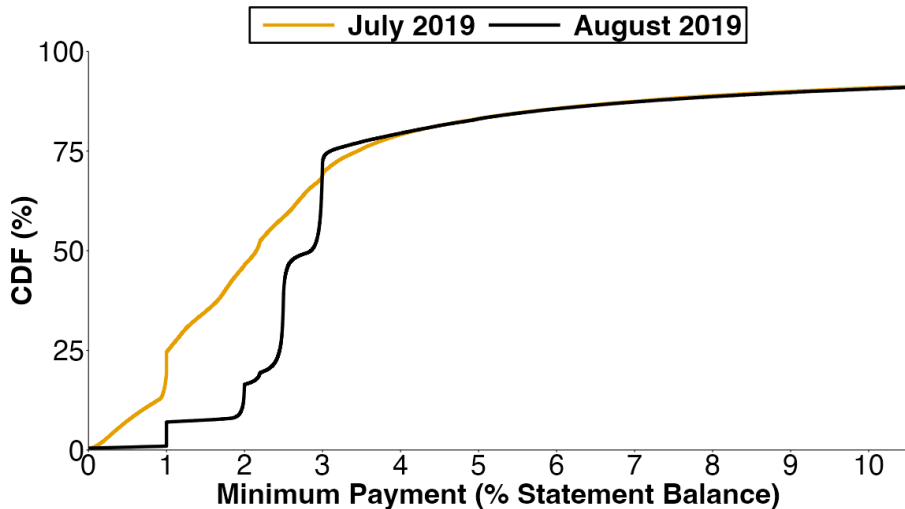
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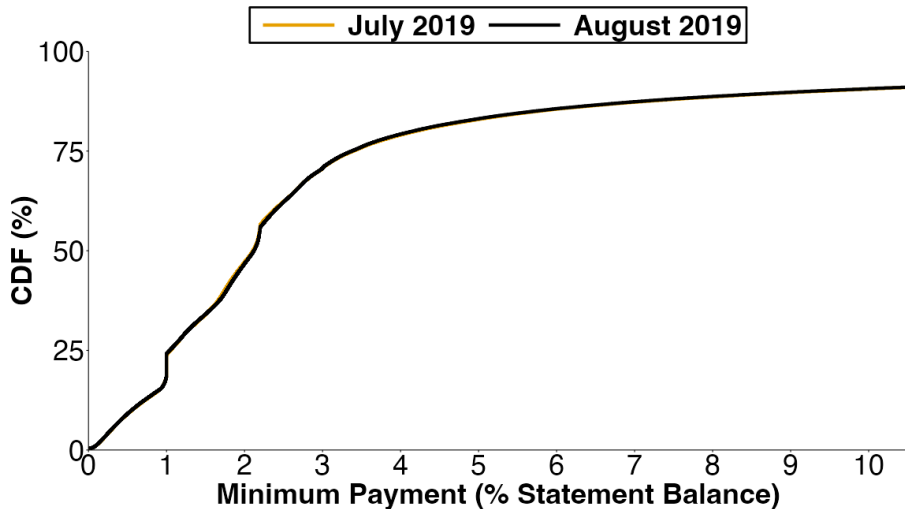
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New cards opened from August 2019: all lenders set minimum payments as:
max {5.0% statement balance, \$10}

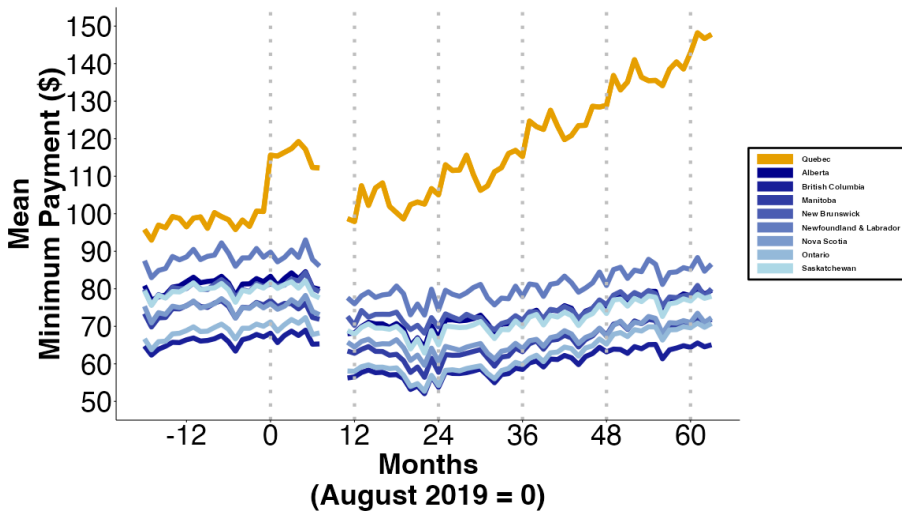
CDF Shows $\uparrow$ Minimum Payments in Quebec in August 2019



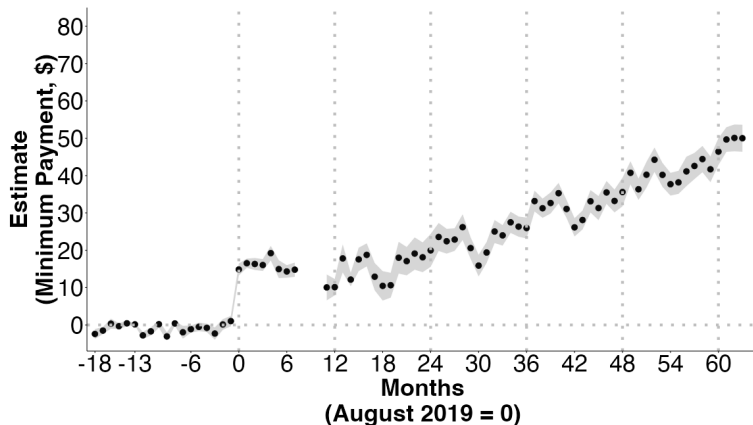
CDF Shows No Change in Minimum Payments in Ontario (or Other Provinces)



Unconditional Means Show $\uparrow$ Minimum Payments in Quebec, No Increases in Minimum Payments in Other Provinces



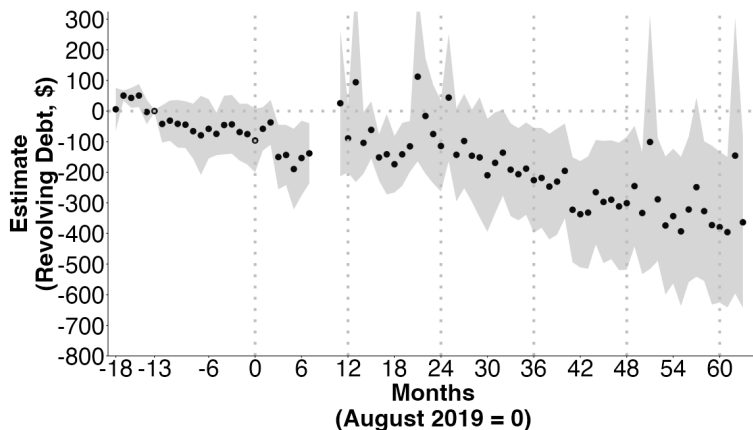
SDID 'First Stage': $\uparrow$ Minimum Payments in Quebec 47% After 60 Months



$t+1$: $\uparrow$ \$16.5 (s.e. \$0.6), $t+6$: $\uparrow$ \$14.3 (s.e. \$0.9), $t+60$: \$46.4 (s.e. \$1.8)

Baseline mean: \$98

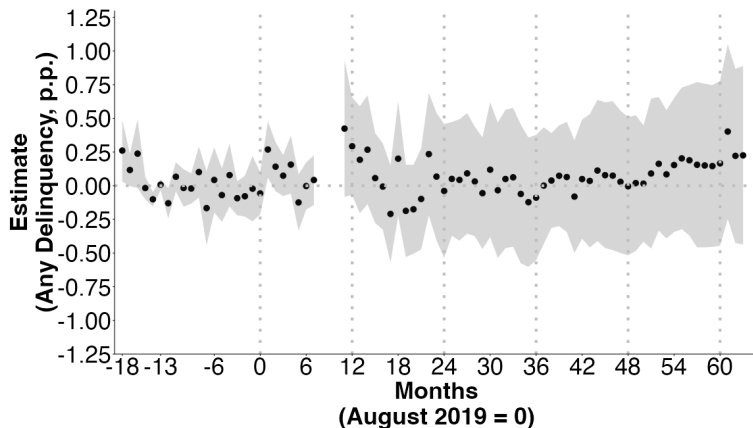
SDID: ↓ Credit Card Revolving Debt 31% After 60 Months (Statement Balance - Actual Payments)



$t+6$: ↓ \$182.8 (s.e. \$61.4), $t+60$: ↓ \$379.3 (s.e. \$125.7)

Baseline mean: \$1,240

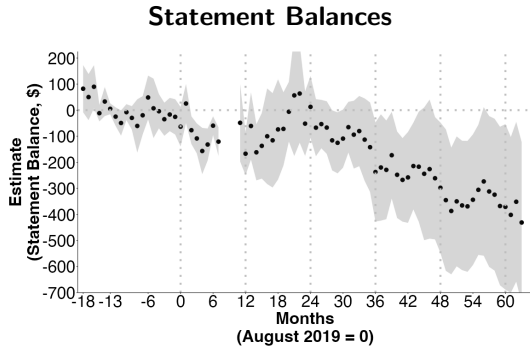
SDID: Effects on Any Credit Card Delinquency (Any 30+ Days Past Due)



t+1: $\uparrow$ 0.2689 percentage points (s.e. 0.1066), t+6: $\downarrow$ 0.0025 (s.e. 0.0890)

Baseline mean: 2.6891 percentage points

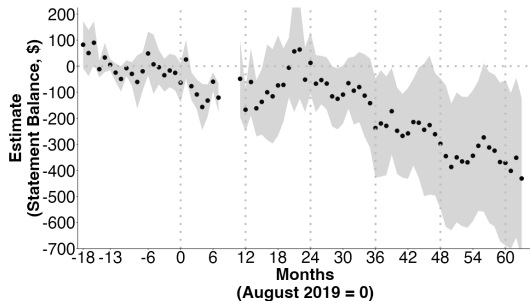
Credit Card Statement Balances ↓ & Credit Limits ↓



Baseline mean: \$3,128

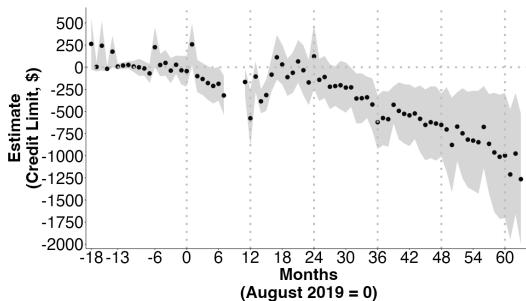
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Statement Balances



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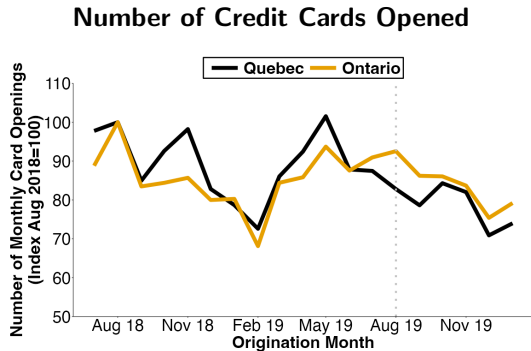
Credit Limits



Baseline mean: \$12,299

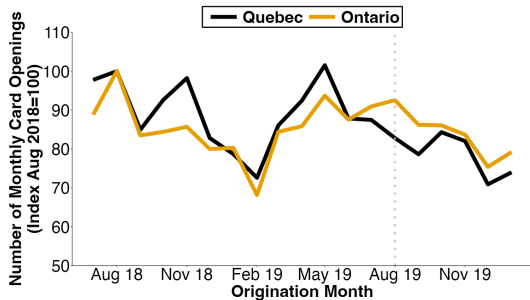
Supply Responses

↓ New Credit Cards Opened, & ↓ Value of New Credit Card Limits in Quebec

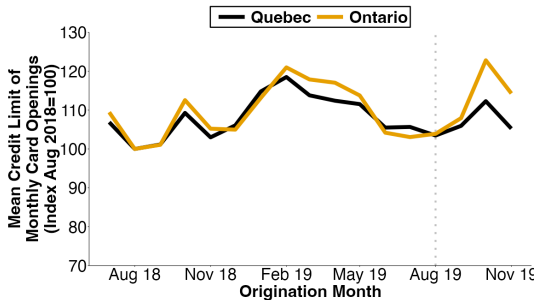


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Number of Credit Cards Opened

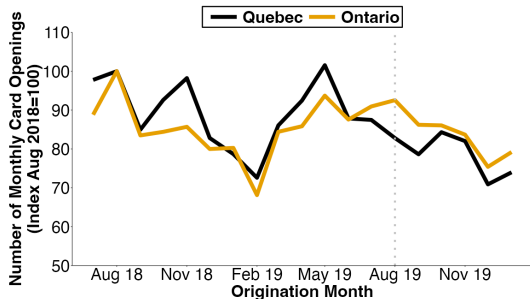


Mean Credit Limits of Cards Opened

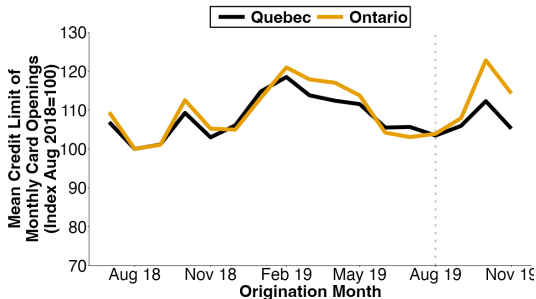


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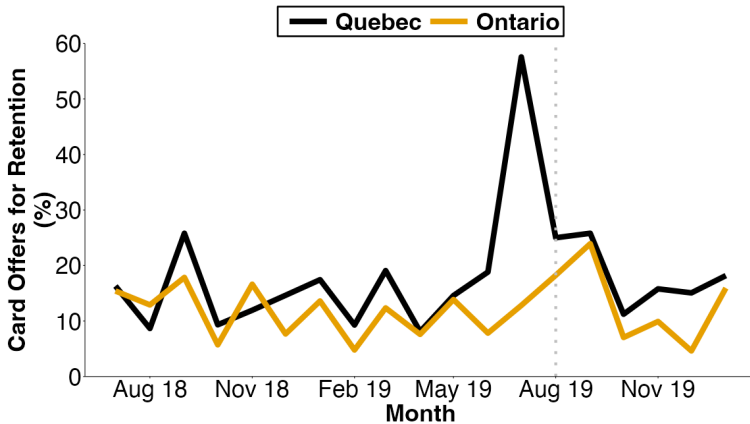


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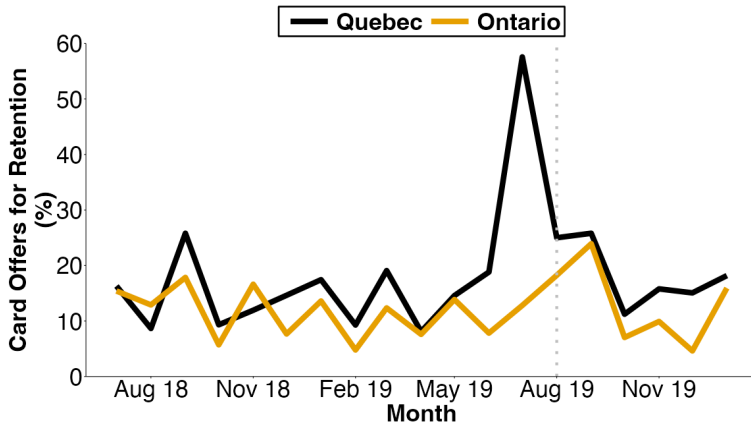


Consistent with higher minimum payments making some consumers unprofitable

Mintel Data Shows Quebec Spike in Mailed Offers to Retain Existing Customers



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Card offers in Quebec just before August 2019 appear to have
↑ credit limits, ↓ APRs, and ↓ annual fees

Conclusions

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- Use comprehensive Canadian credit reporting data and a synthetic difference-in-differences design to estimate the causal effects of a hard paternalistic policy
- Quebec policy permanently ↑ credit card minimum payments by 47% after 5 years
 - Permanently ↓ credit card debt by 31% after 5 years
 - Temporarily ↑ delinquency by 10% for one month
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Next steps

- Heterogeneity by (i) minimum payment exposure, (ii) behavioral consumers
- Spillovers e.g., other debt balances? delinquency? credit scores, autos & mortgages
- Which consumers lose credit card access?
 - revolvers or also transactors? Test of cross-subsidization
 - what happens to consumers without access? Borrow or miss payments elsewhere?
- Selection of new credit cardholders with 5% minimum payments

Thank you!

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SDID: Effects on Transitions into Credit Card Delinquency

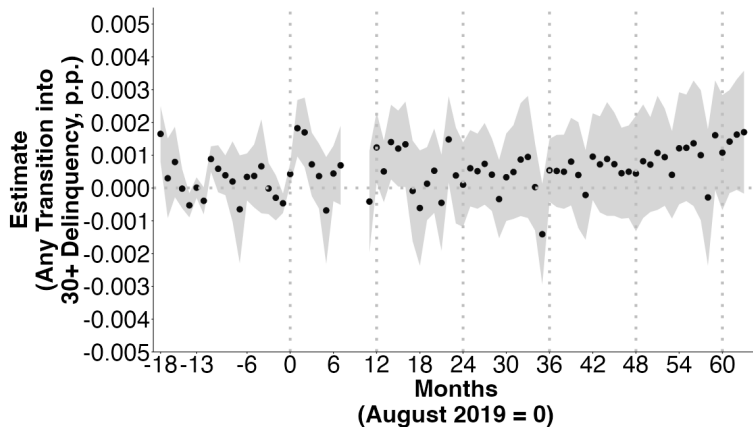


Table 3: Synthetic Difference-in-Differences Estimates for the Effects of the Quebec Policy Over Five Years

Months	Minimum Payment	Revolving Debt	Any Delinquency	Any Delinquency Transition	Statement Balance
0	14.86 (0.82)	-96.21 (53.00)	-0.0571 (0.0795)	0.00043 (0.00017)	-63.43 (38.64)
12	10.13 (1.12)	-89.18 (56.71)	0.2930 (0.1846)	0.00124 (0.00055)	-166.98 (41.96)
24	19.98 (2.24)	-114.16 (75.80)	-0.0389 (0.2535)	0.00010 (0.00066)	12.69 (60.44)
36	25.96 (1.42)	-225.91 (87.50)	-0.0883 (0.2372)	0.00054 (0.00049)	-237.30 (96.93)
48	35.60 (1.95)	-301.04 (110.80)	-0.0042 (0.2613)	0.00044 (0.00090)	-297.68 (127.63)
60	46.43 (1.84)	-379.26 (125.68)	0.1661 (0.3118)	0.00108 (0.00090)	-370.93 (162.02)